1. Subqueries

- A question inside a question
- Put it in (round brackets)
- SQL answers the inner one first

```
SELECT title, pages FROM books
WHERE pages > (SELECT AVG(pages)
FROM books);
```

title	pages
Harry Potter	309
Wonder	310
The Hobbit	310

2. CASE WHEN

- If this, then that
- Rules are checked in order
- ELSE catches everything else; END closes it

```
SELECT title, CASE WHEN pages >= 300
THEN 'Long' ELSE 'Short' END AS size
FROM books;
```

title	size
Harry Potter	Long
Matilda	Short
Wonder	Long
The Hobbit	Long
Holes	Short

3. UNION

- Stack two results on top of each other
- Same number of columns in both
- UNION removes duplicates, UNION ALL keeps them

```
SELECT name FROM members
UNION
SELECT author FROM books;
```

name
Aisha
Ben
Carlos
Dana
Dav Pilkey

4. NULLs and COALESCE

- NULL = missing / unknown
- Find with IS NULL (never = NULL)
- COALESCE fills in a backup value

```
SELECT name,
COALESCE(email, 'no email yet') AS contact
FROM members;
```

name	contact
Aisha	aisha@mail.com
Ben	no email yet
Carlos	carlos@mail.com
Dana	no email yet
Eli	eli@mail.com

5. Practice free + what's next

- sqliteonline.com, db-fiddle.com, sqlime.org (choose SQLite)
- sqlbolt.com for free lessons
- Text and date functions are in the practice guide too
- Next, Part 5: INSERT, UPDATE, DELETE, CREATE TABLE